Problem 4 – ProofCouncil

Abstract

Let (M, ω) be a closed symplectic manifold and $D \subset M$ a symplectic simple normal crossings divisor. After replacing ω by a cohomologous symplectic form joined to it by a path along which D stays simple normal crossings, we construct a tame almost complex structure J such that every point of M has smooth complex coordinates $(z_1, \dots, z_n)$ in which $z_1, \dots, z_r$ are J-holomorphic and $D = \{z_1 \cdots z_r = 0\}$. The proof combines the regularization theorem of Farajzadeh Tehrani, McLean and Zinger with auxiliary approximately holomorphic divisors arranged in a forest of binary trees, whose recursive frames give exact Laurent-monomial transitions between nested systems of local defining functions, and with a convexity property of tame structures that patches local solutions exactly.

1 Introduction

Throughout, (M, ω) is a closed symplectic manifold of real dimension $2n$. A submanifold $V \subset M$ is *symplectic* if $\omega|_V$ is nondegenerate; symplectic submanifolds, and M itself, are oriented by their symplectic forms.

Conventions

Definition 1 (positive symplectic SNC collection)**.** *A finite collection $\{D_i\}_{i \in S}$ of closed embedded symplectic submanifolds of M of real codimension two is a* positive symplectic simple normal crossings (SNC) collection *if for every $\Lambda \subset S$ the* closed intersection $D_\Lambda = \bigcap_{i \in \Lambda} D_i$ *($D_\varnothing = M$) has the following properties: at each point the tangent spaces of the D_i, $i \in \Lambda$, meet transversally, so that D_Λ is a (possibly empty) closed submanifold of codimension $2|\Lambda|$; D_Λ is symplectic; and the orientation of D_Λ as a transverse intersection of the oriented submanifolds D_i inside the oriented manifold M agrees with its symplectic orientation. The union $D = \bigcup_i D_i$ is then a* symplectic SNC divisor *with* branches D_i.

In the terminology of [6], $\{D_i\}$ is an "SC symplectic divisor" and ω belongs to the space $\mathrm{Symp}^+(M, \{D_i\})$ of symplectic forms positive for it. We work with collections rather than with the union D because the proof adjoins auxiliary branches. Since D_Λ has codimension $2|\Lambda|$, at most n branches pass through any point; this elementary fact is used repeatedly.

An almost complex structure J on M is *ω-tame* if $\omega(v, Jv) > 0$ for all $v \neq 0$, and *ω-compatible* if in addition $\omega(Ju, Jv) = \omega(u, v)$. A smooth function $f\colon U \to \mathbb{C}$ on an open set $U \subset M$ is *J-holomorphic* if $df \circ J = i\,df$, equivalently if the complex-linear extension of df vanishes on the $(0,1)$-bundle $T_J^{0,1}M = \{v + iJv\}$. Products, and quotients by nonvanishing functions, of J-holomorphic functions are J-holomorphic. A *smooth complex coordinate system* is a smooth real coordinate system with values in $\mathbb{C}^n$; it need not be a holomorphic chart for J.

Theorem 2. *Let (M, ω) be a closed symplectic manifold and $D = \bigcup_{i \in S} D_i$ a symplectic SNC divisor in the sense of Definition 1. There are a smooth path of symplectic forms ω_t, $0 \leq t \leq 1$, and an ω_1-tame almost complex structure J such that $\omega_0 = \omega$, $[\omega_t] = [\omega] \in H^2(M; \mathbb{R})$ for all t, $\{D_i\}$ is a positive symplectic SNC collection for every ω_t, and every point of M has a neighbourhood with smooth complex coordinates $(z_1, \dots, z_n)$ in which, for some $0 \leq r \leq n$, the functions $z_1, \dots, z_r$ are J-holomorphic and $D = \{z_1 \cdots z_r = 0\}$.*

What is being asked

It is comparatively easy, after the permitted regularization, to find a tame J for which every branch D_i is a J-holomorphic submanifold, $J(TD_i) = TD_i$ (Lemma 3 below; for almost complex structures adapted to a divisor see also [8, Section 3]). Theorem 2 asks for more: a scalar function f with $df \circ J = i\,df$ on an open set and $D_i = \{f = 0\}$ there.

Here is the difficulty. If D_i is the zero set of a transverse section s_i of a line bundle L_i, a local defining function is $f = s_i/F$ for a local nonvanishing frame F. Two frames on overlapping sets differ by a nonvanishing function which must itself be J-holomorphic if both quotients are; unrelated local trivializations have arbitrary transition functions, so no single J serves them all. The central device of the proof is a global system of frames whose transition functions are exact Laurent monomials in other defining functions of the same kind.

Strategy

Regularization (Section 2): by [6, Theorem 2.13], a cohomologous deformation of ω makes the normal bundles of the branches symplectically orthogonal along their intersections; a compatible almost complex structure preserving every branch follows. *Auxiliary divisors* (Section 2): given a line bundle L and a positive SNC collection E, approximately holomorphic sections [5, 1, 9] produce a line bundle H and sections u of H, v of $L \otimes H$ whose zero sets, together with E, form a positive SNC collection (Lemma 4). *The forest* (Section 4): over each branch D_i we grow a full binary tree of depth n; a node a carries (L_a, s_a) and its children carry $L_{a-} = H_a$, $L_{a+} = L_a \otimes H_a$, so that $L_a \cong L_{a+} \otimes L_{a-}^{-1}$. Where $s_{a\pm}$ do not vanish, $s_a/(s_{a+} \otimes s_{a-}^{-1})$ is a defining function of $D_a = s_a^{-1}(0)$; where s_{a+} or s_{a-} is small, that section is in turn replaced by the frame built from its own children, and so on. Since at most n divisors pass through a point, depth n suffices. *Multiscale cover and exact transitions* (Section 5): open sets U_A, indexed by the patterns A of "small" sections, cover

M, and on U_A the recursion gives functions f_a^A with zero sets $D_a \cap U_A$; for $A \subset B$ the f_a^A are Laurent monomials, with coefficient one, in the f_c^B (Lemma 8). *Estimates and local structures* (Section 6): with thresholds $\varepsilon_0 \ll \cdots \ll \varepsilon_n$ attached to the depths, the differentials df_a^B, multiplied by their nonvanishing frames, are uniformly small perturbations of the nearly complex-linear derivatives of the sections s_a, and a pointwise construction gives, near each point, a tame structure for which all functions of a maximal pattern are holomorphic. *Patching* (Sections 3, 7): tame structures are parametrized by strict contractions ("graph maps") relative to a compatible reference structure, and holomorphicity of a fixed function is an *affine* condition on the graph map, so averaging graph maps with a partition of unity preserves every common constraint exactly; Lemma 8 makes the constraints common. The coordinates are read off in Section 8.

Inputs and credit. Besides [6, Theorem 2.13] and the approximately holomorphic techniques of [5, 1, 3, 9], we use the symplectic polar decomposition [10, Theorem 2.16] and Sévennec's description of tame structures as graphs of strict contractions [10, Theorem 2.37]; almost complex structures preserving auxiliary approximately holomorphic hypersurfaces appear in [4, Section 8]. The remaining ingredients, namely the forest of auxiliary divisors, the recursive frames with monomial transitions, the multiscale estimates and the patching with affine constraints, are developed here.

2 The symplectic inputs

In this section (X, σ) is a closed symplectic manifold of dimension $2n$ and $E = \{E_i\}_{i \in S}$ a positive symplectic SNC collection in X (Definition 1), with closed intersections E_Λ, $\Lambda \subset S$. The two lemmas below are consequences of published results.

2.1 Regularization

Farajzadeh Tehrani, McLean and Zinger [6] associate with a symplectic SNC collection the notion of a *regularization*: a compatible system of identifications of neighbourhoods of the strata E_Λ with neighbourhoods of the zero sections of their normal bundles, modelled on split symplectic forms. We use only two consequences: a regularization makes, along each E_Λ, the symplectic normal bundles $N_i = (TE_i)^\sigma|_{E_\Lambda}$, $i \in \Lambda$ (where V^σ denotes the σ-orthogonal complement of V), pairwise σ-orthogonal, so that

$$TX|_{E_\Lambda} = TE_\Lambda \oplus \bigoplus_{i \in \Lambda} N_i \tag{1}$$

is a σ-orthogonal splitting; and [6, Theorem 2.13] (theorem numbers refer to the arXiv version v3), read with its parameter space a point and without relative conditions, provides a path of symplectic forms $\sigma_t = \sigma + d\tau_t$, $\tau_0 = 0$, for which E remains a positive SNC collection and whose endpoint admits a regularization.

Lemma 3 (Regularization)**.** *There is a smooth path $\sigma_t = \sigma + d\tau_t$, $0 \le t \le 1$, of symplectic forms with $\tau_0 = 0$, such that E is a positive symplectic SNC collection for every σ_t and such that for σ_1 the splittings* (1) *(with $N_i = (TE_i)^{\sigma_1}$) are σ_1-orthogonal. Moreover σ_1 admits a compatible almost complex structure K with $K(TE_i) = TE_i$ for every i.*

Proof. The path is [6, Theorem 2.13], as just explained. For the compatible structure, recall the symplectic polar decomposition [10, proof of Theorem 2.16]: a Riemannian metric g determines a g-skew endomorphism A by $\sigma_1(u,v) = g(Au,v)$, and $J_g = A(A^*A)^{-1/2}$ is σ_1-compatible. If a subspace $W \subset T_xX$ satisfies $A(W) \subset W$, then also $A^*(W) = A(W) \subset W$, hence $(A^*A)^{-1/2}$, a limit of polynomials in A^*A, preserves W, and so does J_g. For $W = T_xE_i$ one has $A(W) \subset W$ exactly when $g(Au,v) = \sigma_1(u,v) = 0$ for all $u \in W$ and $v \in W^{\perp_g}$, that is, when the g-orthogonal complement of T_xE_i is its σ_1-orthogonal complement $N_i(x)$. So it suffices to find a metric g such that

$$T_xE_i \perp_g N_i(x) \qquad \text{for all } i \text{ and all } x \in E_i. \tag{2}$$

For fixed x and i this is a linear condition on g (orthogonality of two fixed complementary subspaces), so the metrics on T_xX satisfying (2) for all i with $x \in E_i$ form a convex set. At a point $x \in E_\Lambda$ lying on no deeper stratum this set consists exactly of the metrics making (1) orthogonal at x, because $T_xE_i = T_xE_\Lambda \oplus \bigoplus_{j \in \Lambda\setminus\{i\}} N_j(x)$; and a metric orthogonalizing the finer splitting of a deeper stratum $E_{\Lambda''} \ni x$, $\Lambda'' \supset \Lambda$, also orthogonalizes (1). Since (2) concerns only points of $\bigcup_i E_i$, define g on $TX|_{E_\Lambda}$ by descending induction on $|\Lambda|$: the metric already defined over the deeper strata contained in E_Λ orthogonalizes (1) there; extend it to a neighbourhood in E_Λ summand by summand, and combine it with local metrics orthogonalizing (1) by a partition of unity on E_Λ, using convexity. Finally extend g to X; then $K = J_g$ is the required structure. □

2.2 Auxiliary divisors

We recall the framework of approximately holomorphic sections. Suppose $[\sigma/(2\pi)]$ is integral, let $A \to X$ be a Hermitian line bundle with unitary connection of curvature $-i\sigma$, let K be σ-compatible and $g = \sigma(\cdot, K\cdot)$. Sections of A^k, or of $F \otimes A^k$ for a fixed Hermitian vector bundle F, are measured in the rescaled metric $g_k = kg$, in which the curvature of A^k has unit size. A family s_k is *approximately holomorphic* if s_k and its first few covariant derivatives are bounded in g_k-norm while the K-antilinear part $\bar\partial_K s_k$ of ∇s_k and its derivatives are $O(k^{-1/2})$; it is *η-transverse to zero* along a compact submanifold $Y \subset X$ if at every $y \in Y$ with $|s_k(y)| < \eta$ the derivative $\nabla s_k|_{T_yY}$ is surjective with a right inverse of g_k-norm at most η^{-1}. Donaldson [5] constructed, for all large k, approximately holomorphic sections of A^k that are η-transverse along $Y = X$ with $\eta > 0$ independent of k; Auroux [1, Theorem 2 and Corollary 1] extended this to $F \otimes A^k$; Mohsen's relative theorem [9, p. 657] achieves η-transversality of the restriction to an arbitrary compact submanifold Y, not necessarily K-invariant, and its twisted version [9, p. 662] allows a fixed F. The sections are obtained by perturbations of controlled size, so finitely many such conditions can be imposed successively, each later

perturbation being smaller than the transversality bounds already secured (Donaldson's globalization principle, see [3, Proposition 4.1]).

Transversality produces symplectic submanifolds because of the following linear-algebra fact, due to Donaldson [5]: if at a zero y of s the K-complex-linear part of ∇s is surjective with right inverse of norm at most $2\eta^{-1}$ and the antilinear part is small compared with η (both hold when ∇s is η-transverse and the antilinear part is at most $\eta/2$), then $\ker \nabla s(y)$ is the graph of a small map over a complex subspace, hence a symplectic subspace whose orientation agrees with the complex orientation of that subspace. The same holds for the restriction to a K-invariant submanifold Y, within (T_yY, K). Since complex orientations are compatible with transverse intersection, all intersections obtained in this way are positively oriented in the sense of Definition 1.

Lemma 4 (Auxiliary divisors)**.** *Let (X, σ) be closed, E a finite positive symplectic SNC collection in X, and $L \to X$ a smooth complex line bundle. After a deformation of σ as in Lemma 3 there are a complex line bundle H and smooth sections $u \in \Gamma(H)$, $v \in \Gamma(L \otimes H)$, transverse to the zero section, such that $E \cup \{u^{-1}(0), v^{-1}(0)\}$ is a positive symplectic SNC collection. Moreover the orientations induced on $u^{-1}(0)$ and $v^{-1}(0)$ by the sections u, v (through the complex orientations of their bundles) are the symplectic ones.*

Proof. Apply Lemma 3; denote the resulting form again by σ and let K be the compatible structure preserving every E_i, hence every closed intersection E_Λ; put $g = \sigma(\cdot, K\cdot)$. We work with the compact E_Λ rather than with the open strata because the transversality theorems require compactness.

Step 1: the integral case. Suppose $[\sigma/(2\pi)]$ is integral, let A have curvature $-i\sigma$ and put $H = A^k$. The pair (u, v) is a section of $(\underline{\mathbb{C}} \oplus L) \otimes A^k$. For every nonempty E_Λ, including $E_\varnothing = X$, impose η-transversality to zero along E_Λ for each of u, v and (u, v), using the relative twisted theorem of [9]. These are finitely many conditions, achieved successively as explained above, with a common $\eta > 0$ for all large k. Transversality of $u|_{E_\Lambda}$ gives $u^{-1}(0) \pitchfork E_\Lambda$; transversality of $(u, v)|_{E_\Lambda}$ gives $u^{-1}(0) \pitchfork v^{-1}(0)$ along E_Λ and $u^{-1}(0) \cap v^{-1}(0) \pitchfork E_\Lambda$. Where the expected dimension of an intersection is negative, η-transversality means $|s| \geq \eta$ on E_Λ, so the intersection is empty. Since E_Λ is K-invariant, $\nabla(s|_{E_\Lambda})$ splits into K-linear and K-antilinear parts, and the antilinear part is $O(k^{-1/2})$ by approximate holomorphicity; Donaldson's linear-algebra fact therefore shows that every new intersection is symplectic and positively oriented, so $E \cup \{u^{-1}(0), v^{-1}(0)\}$ is a positive SNC collection. The same fact gives the orientation statement: deforming ∇s linearly to its complex-linear part through surjective maps, the orientation induced by s on its zero set is the complex orientation of the nearby complex kernel, which is the symplectic orientation.

Step 2: removing the integrality assumption. Call a symplectic form σ' *rational* if $[\sigma'/(2\pi)] \in H^2(X; \mathbb{Q})$. In general σ is not rational, and we follow the device of [2, Section 1]: a nearby rational form is used only to construct the polarizing bundle, while symplecticity is checked for σ itself. Fix closed 2-forms α_j representing a basis of $H^2(X; \mathbb{R})$ and consider the forms $\sigma' = \sigma + \sum_j c_j \alpha_j$ with c in a small closed ball; all are symplectic with uniformly bounded derivatives, and rational forms are dense among them. Polar decomposition with respect to the fixed metric $g = \sigma(\cdot, K\cdot)$ gives σ'-compatible structures K' depending

smoothly on σ', with $K' = K$ at $\sigma' = \sigma$. For a rational σ' choose a positive integer q with $q[\sigma'/(2\pi)]$ integral, let A have curvature $-iq\sigma'$ and again set $H = A^k$. Since the curvature of A^k is $-iN\sigma'$ with $N = qk$, the effective large parameter is N and the effective metric is $g_N = Ng'$, $g' = \sigma'(\,\cdot\,, K'\cdot\,)$.

Step 1, carried out for (σ', K') and the parameter N, provides a transversality constant η for the finitely many restriction conditions. The point requiring attention is that η must not depend on the denominator q, because the rational approximation will be chosen in terms of η. Uniformity in q is obtained from the quantitative construction, not merely from qualitative openness; it is a deduction from the structure of the cited estimates, which we indicate. Mohsen's linear restriction estimate has constants depending only on dimensions and radii [9, Theorem 6.5 and Corollary 6.6]. In local unitary gauges at length scale $N^{-1/2}$ the curvature $-iN\sigma'$ is normalized by N, so the geometric bounds required by the construction are uniform over the compact family of σ', and the fixed twist L contributes only lower-order terms. The local perturbation and globalization estimates [9, Proposition 8.2 and Sections 8.3–8.6] use these bounds together with counts of separated sets in the fixed compact E_Λ. They therefore give a common bound $\eta > 0$, independent of q, for all the finitely many conditions when N is large.

The relative theorem does not require K' to preserve E_Λ, but K does preserve E_Λ. For $s = u, v, (u, v)$ the derivative along E_Λ therefore satisfies, in the rescaled metrics,

$$\big\|(\nabla s|_{TE_\Lambda})^{0,1}_K\big\|_{g_N} \le C\|K' - K\|_{C^0} + CN^{-1/2},$$

the first term coming from the difference between the K'- and K-antilinear parts and the second from approximate K'-holomorphicity. Choose the rational approximation σ' so close to σ that the first term is small compared with η, then take k large to control the second. Donaldson's linear-algebra fact now applies with respect to K, which is σ-compatible: all new intersections are σ-symplectic and positively oriented, not merely σ'-symplectic.

The rational form is construction data only and is not inserted into the deformation path. The sole change of σ in this lemma is the initial regularization deformation of Lemma 3. □

3 Two linear-algebra facts about tame structures

For an almost complex structure J on a symplectic manifold (X, σ) write $T^{1,0}_J X$ and $T^{0,1}_J X$ for the $\pm i$ eigenbundles of J on $TX \otimes \mathbb{C}$. The first fact is Sévennec's parametrization of tame structures by strict contractions [10, Theorem 2.37], together with the observation that holomorphicity constraints are affine in this parametrization.

Lemma 5 (Convex patching with holomorphic-function constraints)**.** *Let K be a σ-compatible almost complex structure on X and $E^+ = T^{1,0}_K X$, $E^- = T^{0,1}_K X$. Every σ-tame almost complex structure J has $T^{0,1}_J X = \{\xi + \Phi_J\xi : \xi \in E^-\}$ for a unique complex-linear bundle map $\Phi_J \colon E^- \to E^+$ (the* graph map *of J) with $\|\Phi_J\| < 1$ pointwise, and conversely every such Φ arises from a unique tame J. For a smooth function f, the condition that f be J-holomorphic*

is the affine linear condition

$$df(\xi + \Phi_J \xi) = 0 \qquad (\xi \in E^-) \tag{3}$$

on Φ_J. Consequently, if J_α are σ-tame structures on open sets W_α, (ρ_α) is a partition of unity subordinate to (W_α), and J is the tame structure with $\Phi_J = \sum_\alpha \rho_\alpha \Phi_{J_\alpha}$, then every function that is J_α-holomorphic at y for all α with $\rho_\alpha(y) \neq 0$ is J-holomorphic at y.

Proof. On $TX \otimes \mathbb{C}$ consider the Hermitian form $\mathcal{H}(\xi, \eta) = -i\,\sigma(\xi, \overline{\eta})$. The summands E^+ and E^- are $\mathcal{H}$-orthogonal, $\mathcal{H}$ is positive definite on E^+ and negative definite on E^-, and we use these restrictions (with the sign reversed on E^-) as Hermitian metrics to define $\|\Phi\|$. For $\xi = v + iJv \in T_J^{0,1}X$ one computes $\mathcal{H}(\xi, \xi) = -2\sigma(v, Jv)$, so for tame J the bundle $T_J^{0,1}X$ is $\mathcal{H}$-negative definite of complex rank n. Such a subspace meets the positive definite E^+ trivially, hence is the graph of a complex-linear map $\Phi_J \colon E^- \to E^+$, and negativity of $\mathcal{H}(\xi + \Phi\xi, \xi + \Phi\xi) = -|\xi|^2 + |\Phi\xi|^2$ says exactly $\|\Phi_J\| < 1$. Conversely, the graph of a strict contraction Φ is negative definite, its complex conjugate is positive definite, the two are complementary, and they are the $(0,1)$- and $(1,0)$-bundles of a real almost complex structure J; the computation above, read backwards, shows that J is tame.

Condition (3) is the vanishing of df on $T_J^{0,1}X$ and is affine in Φ_J. A convex combination of strict contractions is a strict contraction, and a convex combination of solutions of the affine equations (3) for a fixed f is again a solution. □

Averaging the endomorphisms J_α themselves would not produce an almost complex structure; averaging graph maps does, and no derivative of the partition of unity enters, so the holomorphicity statements are exact.

Lemma 6 (Graph construction)**.** *Let I be a σ-tame almost complex structure, $V \to X$ a complex vector bundle with a Hermitian metric, and λ a V-valued one-form on an open set $W \subset X$, with complex-linear extension to $TX \otimes \mathbb{C}$ restricting to*

$$P = \lambda|_{T_I^{1,0}X} \colon T_I^{1,0}X \to V, \qquad Q = \lambda|_{T_I^{0,1}X} \colon T_I^{0,1}X \to V.$$

Suppose P is surjective on W and let R be a smooth right inverse of P, for instance $R = P^*(PP^*)^{-1}$, which has the least norm among all right inverses of P. Put

$$\mu = -RQ \colon T_I^{0,1}X \to T_I^{1,0}X, \qquad E_\mu = \{\xi + \mu\xi : \xi \in T_I^{0,1}X\}. \tag{4}$$

Then λ vanishes on E_μ. If X is compact and a fixed metric is used, there is $\mu_0 > 0$ such that whenever $\|\mu\| < \mu_0$ on W, the bundle E_μ is the $(0,1)$-bundle of a σ-tame almost complex structure J_μ on W, depending smoothly on the data. In particular, if $V = \bigoplus_a L_a$ and the components of λ have the form $\lambda_a = F_a\, df_a$ with F_a a nonvanishing section of L_a over W and f_a a smooth function, then every f_a is J_μ-holomorphic on W.

Proof. For $\xi \in T_I^{0,1}X$, $\lambda(\xi + \mu\xi) = Q\xi + P\mu\xi = Q\xi - PRQ\xi = 0$. With $\mathcal{H}$ as in the previous proof, $T_I^{0,1}X$ is $\mathcal{H}$-negative definite because I is tame, and negativity is an open condition; on a compact manifold there is a uniform μ_0 such that E_μ is negative definite whenever $\|\mu\| < \mu_0$. Then E_μ and $\overline{E_\mu}$ are complementary and define a tame almost complex structure J_μ with $T_{J_\mu}^{0,1}X = E_\mu$, exactly as in the proof of Lemma 5. Finally, $\lambda_a = F_a\, df_a$ vanishes on E_μ and F_a is nonvanishing, so df_a vanishes on $E_\mu = T_{J_\mu}^{0,1}X$. □

4 Building the finite forest

From now on $X = M$ with the given divisor $D = \bigcup_{i \in S} D_i$, where $S = \{1, \dots, m\}$. If $D = \varnothing$, Theorem 2 holds with $\omega_t = \omega$, any tame J and $r = 0$, so assume $D \neq \varnothing$.

Divisor bundles of the branches

For every branch D_i choose a smooth complex line bundle $L_i \to X$ with a section s_i transverse to the zero section whose zero set is D_i, the orientation induced on D_i by s_i (through the complex orientation of L_i) being the symplectic one: pull the oriented normal bundle of D_i back to a tubular neighbourhood, take the tautological section there, and glue to the trivial bundle with the constant section 1 over $X \setminus D_i$, using that the tautological section does not vanish on the overlap (compare [7, Section 2.2]).

The trees

Make each $i \in S$ the root of a full binary tree of depth n. A node is a word $a = iw$ with $w \in \{+, -\}^d$, $0 \leq d \leq n$; its depth is $d(a) = d$, roots have depth 0, *leaves* have depth n, and a non-leaf node a has the two children $a+$ and $a-$. Let $\mathcal{T}$ be the finite set of all nodes of all m trees. We attach to every node a a line bundle L_a with a transverse section s_a, and to every non-leaf node a bundle isomorphism θ_a, by induction on depth. Suppose L_a, s_a have been constructed for all nodes of depth $\leq d < n$, with the zero sets $D_a = s_a^{-1}(0)$ of all nodes constructed so far forming a positive SNC collection for the current symplectic form. Process the nodes a of depth d one at a time: apply Lemma 4 with $L = L_a$ and E the collection of all divisors constructed so far (including those produced for earlier nodes of depth d), and set

$$L_{a-} = H_a, \quad s_{a-} = u, \qquad L_{a+} = L_a \otimes H_a, \quad s_{a+} = v.$$

Retain the canonical complex bundle isomorphism

$$\theta_a \colon L_{a+} \otimes L_{a-}^{-1} = L_a \otimes H_a \otimes H_a^{-1} \xrightarrow{\cong} L_a. \tag{5}$$

Here and below inverse line bundles and inverses of nonvanishing sections are taken in the tensor sense: F^{-1} is the section of $L^{-1} = L^*$ dual to the nonvanishing section F of L. Each application of Lemma 4 includes a regularization deformation of the symplectic form, which preserves all divisors constructed so far as a positive SNC collection. There are $m(2^n - 1)$ applications. At the end, the finite collection $\{D_a\}_{a \in \mathcal{T}}$ is a positive symplectic SNC collection for a symplectic form cohomologous to ω, obtained from ω by concatenating finitely many paths as in Lemma 3.

Apply Lemma 3 once more to the full collection $\{D_a\}_{a \in \mathcal{T}}$. Denote the resulting symplectic form by σ and let I be the σ-compatible (in particular σ-tame) almost complex structure provided by the lemma, which preserves every D_a. Thus I preserves TD_a and induces a complex structure on each normal line $N_x D_a = T_x X / T_x D_a$.

Normalizing the sections along their zero sets

Along D_a the section s_a vanishes, so its derivative there is an intrinsic L_a-valued one-form $ds_a\colon TX|_{D_a} \to L_a|_{D_a}$, independent of any connection, with kernel TD_a. We arrange, by changing the sections but not their bundles or zero sets, that

$$(ds_a) \circ I = i\, ds_a \quad \text{on } D_a \qquad (a \in \mathcal{T}), \tag{6}$$

that is, ds_a is complex linear along D_a. Indeed, at $x \in D_a$ the derivative descends to a real-linear isomorphism $N_x D_a \to (L_a)_x$ between complex lines, which may be written $z \mapsto Az + B\bar{z}$ with $A, B \in \mathbb{C}$. It is orientation preserving: the orientation of D_a induced by s_a is the symplectic one (by construction for roots, by Lemma 4 for the other nodes; both orientations are unchanged along the subsequent deformations), and since I is tame and preserves TD_a, the symplectic orientations of X and D_a are the complex orientations of (TX, I) and (TD_a, I), so that $N_x D_a$ carries the quotient complex orientation. Hence $|A| > |B|$, and by compactness $|A| - |B|$ is bounded below on D_a. Identify a tubular neighbourhood of D_a with a neighbourhood of the zero section of ND_a and identify L_a there with the pull-back of $L_a|_{D_a}$. In these terms $s_a = Az + B\bar{z} + O(|z|^2)$, and we replace s_a by $s_a - \chi B\bar{z}$, where χ is a cutoff equal to 1 near D_a and supported in a small tubular neighbourhood. During this change the section has the form $Az + (1-\chi)B\bar{z} + O(|z|^2)$, whose first-order part has modulus at least $(|A| - |B|)|z|$; so for a sufficiently small neighbourhood no new zeros appear and transversality is preserved. The new normal derivative is $z \mapsto Az$, complex linear. Perform this independently for all nodes; the isomorphisms (5) are unaffected.

From this point on σ, I and all the data $(L_a, s_a, \theta_a, D_a)_{a\in\mathcal{T}}$ are fixed.

5 Recursive frames and their transition functions

Where $s_{a\pm}$ do not vanish, $F = \theta_a(s_{a+} \otimes s_{a-}^{-1})$ is a frame of L_a and $f = s_a/F$ is a defining function of D_a; nearer to D_{a+} or D_{a-} this frame degenerates and the small child section must itself be replaced by the frame built from its own children. The bookkeeping device is an *active set*: the set of nodes whose sections are "small" at the points under consideration, at depth-dependent thresholds chosen so that a section is always much smaller than the sections framing it.

Active sets and the multiscale cover

Call a set A of non-leaf nodes *active* if it is ancestor-closed: whenever a node belongs to A, all its ancestors do. Its *boundary* ∂A consists of the roots not in A together with the children of nodes of A that do not belong to A (these children may be leaves). The empty set is active, with $\partial\varnothing = S$. There are finitely many active sets.

Choose Hermitian metrics on all L_a. We shall fix constants $0 < \varepsilon_0 < \varepsilon_1 < \cdots < \varepsilon_n$ in Section 6; they will be small, with each ratio $\varepsilon_d/\varepsilon_{d+1}$ small. For an active set A define the

open set

$$U_A = \Big\{ x \in X : \ |s_a(x)| < 2\varepsilon_{d(a)} \text{ for } a \in A, \quad |s_b(x)| > \varepsilon_{d(b)} \text{ for } b \in \partial A \Big\}. \tag{7}$$

Since $\{D_a\}_{a\in\mathcal{T}}$ is a positive SNC collection in dimension $2n$, no $n+1$ of the divisors D_a have a common point, and by compactness there is $\delta_0 > 0$ such that no $n+1$ distinct sections s_a have norm $\le \delta_0$ at the same point.

Lemma 7 (The cover)**.** *If $2\varepsilon_n < \delta_0$, then every nonempty U_A has $|A| \le n$, and the sets U_A, A active, cover X.*

Proof. If $x \in U_A$ then $|s_a(x)| < 2\varepsilon_{d(a)} \le 2\varepsilon_n < \delta_0$ for all $a \in A$, so $|A| \le n$. For the covering, fix x and determine A by the following process. Start at each root; call a node *accessible* if it is a root or the child of an activated node; activate an accessible node a precisely when $|s_a(x)| \le \varepsilon_{d(a)}$, and continue to its children only if it is activated. An accessible leaf ℓ is never activated: otherwise ℓ and its n ancestors would be $n+1$ distinct nodes with sections of norm $\le \varepsilon_n < \delta_0$ at x. Hence the set A of activated nodes is an ancestor-closed set of non-leaf nodes, and its boundary consists exactly of the accessible nodes that were not activated, at which $|s_b(x)| > \varepsilon_{d(b)}$. So $x \in U_A$. □

Frames and functions

For an active set A let $O_A = \{x \in X : s_b(x) \neq 0 \text{ for all } b \in \partial A\}$, an open set containing $\overline{U_A}$, since $|s_b| \ge \varepsilon_{d(b)} > 0$ on $\overline{U_A}$ for $b \in \partial A$. On O_A define nonvanishing sections F_c^A of L_c, for $c \in A \cup \partial A$, recursively from the boundary upwards, together with functions f_a^A for $a \in A$:

$$F_b^A = s_b \qquad (b \in \partial A), \tag{8}$$

$$F_a^A = \theta_a\big(F_{a+}^A \otimes (F_{a-}^A)^{-1}\big) \qquad (a \in A),$$

$$f_a^A = s_a / F_a^A \qquad (a \in A). \tag{9}$$

The recursion is well defined because for $a \in A$ each child of a lies in $A \cup \partial A$, and the frames are nonvanishing by induction. Each f_a^A is an ordinary smooth complex-valued function on O_A, and its zero set is $D_a \cap O_A$.

Lemma 8 (Exact monomial transitions)**.** *Let $A \subset B$ be active sets. Then for every $a \in A$ there are integers k_b, $b \in \partial A \cap B$, such that*

$$f_a^A = f_a^B \prod_{b \in \partial A \cap B} (f_b^B)^{k_b} \qquad \text{on } O_A \cap O_B, \tag{10}$$

where the functions f_b^B, $b \in \partial A \cap B$, are nonvanishing on $O_A \cap O_B$, in particular on $U_A \cap O_B$.

Proof. First, every $b \in \partial A$ lies in $B \cup \partial B$: if $b \notin B$, then b is either a root not in B or a child of a node of $A \subset B$ not in B, hence $b \in \partial B$. Therefore F_c^B is defined for all $c \in A \cup \partial A$, and we may consider the ratios

$$r_c = F_c^A / F_c^B \in C^\infty(O_A \cap O_B, \mathbb{C}^*) \qquad (c \in A \cup \partial A),$$

quotients of two nonvanishing sections of the same line bundle L_c. We show by induction from the boundary upwards that each r_c is a Laurent monomial with coefficient one in the f_b^B, $b \in \partial A \cap B$. For $b \in \partial A$: if $b \notin B$ then $b \in \partial B$ and both constructions use s_b as the frame, so $r_b = 1$; if $b \in B$ then $r_b = s_b/F_b^B = f_b^B$, which is therefore nonvanishing on $O_A \cap O_B$. For $a \in A$, complex linearity of θ_a gives

$$F_a^A = \theta_a\big(r_{a+}F_{a+}^B \otimes r_{a-}^{-1}(F_{a-}^B)^{-1}\big) = \frac{r_{a+}}{r_{a-}}\,F_a^B, \qquad \text{i.e.} \qquad r_a = \frac{r_{a+}}{r_{a-}};$$

the isomorphism θ_a cancels exactly. Finally

$$f_a^A = \frac{s_a}{F_a^A} = \frac{s_a}{F_a^B}\cdot\frac{F_a^B}{F_a^A} = f_a^B\, r_a^{-1},$$

which is (10). □

By (10), for every almost complex structure J, if all the functions f_c^B occurring on the right-hand side are J-holomorphic, then so is f_a^A; a transition $g\,f_a^B$ with an arbitrary nonvanishing smooth g would not have this property, which is why actual bundle isomorphisms are used rather than unrelated local trivializations.

6 Uniform estimates and local tame structures

The goal of this section is Proposition 10: near any point where an active set B is "small and its boundary is not", there is a tame structure making all f_a^B, $a \in B$, holomorphic. By Lemma 6 it suffices that the one-forms $F_a^B\,df_a^B$ have jointly surjective $(1,0)$-parts with uniformly bounded right inverses and small $(0,1)$-parts. All constants are fixed *before* the thresholds ε_d.

Fix Hermitian connections ∇_a on the bundles L_a and an I-Hermitian metric on TX. All norms below refer to these fixed metrics. For an L_a-valued one-form α we write $\alpha_I^{1,0}$ and $\alpha_I^{0,1}$ for its restrictions to $T_I^{1,0}X$ and $T_I^{0,1}X$, so that $\alpha_I^{0,1} = 0$ means that α is I-complex linear.

Lemma 9 (Uniform facts)**.** *There are $\delta > 0$ and $C_0, C_1 > 0$ such that:*

1. *No $n+1$ distinct sections s_a have norm $\le \delta$ at a common point.*
2. *Whenever $\Lambda \subset \mathcal{T}$ and $x \in X$ satisfy $|s_a(x)| \le \delta$ for all $a \in \Lambda$, the complex-linear map*

$$P_\Lambda = \big((\nabla_a s_a)_I^{1,0}\big)_{a\in\Lambda} \colon T_{I,x}^{1,0}X \longrightarrow \bigoplus_{a\in\Lambda}(L_a)_x \tag{11}$$

   is surjective with a right inverse of norm at most C_0.

3. *For every node a,*

$$|(\nabla_a s_a)_I^{0,1}| \le C_1|s_a|, \qquad |\nabla_a s_a| \le C_1 \quad \text{on } X. \tag{12}$$

Proof. (1) holds for every $\delta \le \delta_0$, by the choice of δ_0 preceding Lemma 7. For (2), fix Λ with $|\Lambda| \le n$ (by (1), larger Λ do not occur). On the compact common zero set $D_\Lambda = \bigcap_{a\in\Lambda} D_a$ the connection derivatives $\nabla_a s_a$ coincide with the intrinsic derivatives ds_a, which are I-complex linear by (6) and jointly surjective onto $\bigoplus_{a\in\Lambda} L_a$ by the transversality of the D_a; so P_Λ is a surjective complex-linear map there, with a right inverse bounded by compactness. Surjectivity with a bound is an open condition, so it persists on a neighbourhood of D_Λ, and by compactness this neighbourhood contains $\{|s_a| \le \delta \ \forall a \in \Lambda\}$ for small δ; if $D_\Lambda = \varnothing$, this set is empty for small δ. There are finitely many Λ. For (3), the second bound is compactness. For the first, $(\nabla_a s_a)^{0,1}_I$ vanishes on D_a by (6), so it is bounded by a constant times the distance to D_a, and transversality of s_a makes this distance comparable to $|s_a|$ near D_a; away from D_a, $|s_a|$ is bounded below. □

Differentials of the recursive frames

Let B be an active set and consider a point where

$$|s_a| \le 2\varepsilon_{d(a)} \quad (a \in B), \qquad |s_b| \ge \varepsilon_{d(b)} \quad (b \in \partial B); \tag{13}$$

these are the closed versions of the inequalities defining U_B, and every point of $\overline{U_B}$ satisfies them. Such a point lies in O_B, so the frames F^B_c are defined near it. Their logarithmic derivatives $\beta^B_c = (F^B_c)^{-1}\nabla_c F^B_c$, $c \in B \cup \partial B$, are ordinary complex-valued one-forms. For each non-leaf node a, the connection on L_a transported by θ_a from the tensor-product connection on $L_{a+} \otimes L_{a-}^{-1}$ differs from ∇_a by a fixed smooth complex one-form γ_a, bounded on X. Hence the recursion (8) gives

$$\begin{aligned} \beta^B_a &= \beta^B_{a+} - \beta^B_{a-} + \gamma_a && (a \in B), \\ \beta^B_b &= s_b^{-1}\nabla_b s_b && (b \in \partial B). \end{aligned}$$

Expanding the recursion, β^B_a is a signed sum of boundedly many forms γ_c and of terminal terms $s_b^{-1}\nabla_b s_b$ with b a boundary descendant of a. If a has depth d, all these boundary descendants have depth at least $d+1$, so at a point satisfying (13) the terminal terms are bounded by $C_1/\varepsilon_{d(b)} \le C_1/\varepsilon_{d+1}$ by (12). Assuming $\varepsilon_n \le 1$, we obtain a constant C_2, depending only on C_1, the forest and the γ_c, with

$$|\beta^B_a| \le C_2/\varepsilon_{d(a)+1} \qquad (a \in B). \tag{14}$$

Now multiply the differential of f^B_a by its frame. Since $s_a = f^B_a F^B_a$, the Leibniz rule gives $\nabla_a s_a = F^B_a\, df^B_a + f^B_a F^B_a \beta^B_a$, that is, the L_a-valued one-form

$$\lambda^B_a := F^B_a\, df^B_a = \nabla_a s_a - s_a \beta^B_a \tag{15}$$

satisfies, by (13) and (14), at points satisfying (13),

$$|\lambda^B_a - \nabla_a s_a| \le 2C_2\, \frac{\varepsilon_{d(a)}}{\varepsilon_{d(a)+1}} \qquad (a \in B), \tag{16}$$

the size of s_a divided by the size of the sections framing it.

Choice of the thresholds

Choose

$$\varepsilon_d = t^{\,n-d}\varepsilon_n \qquad (0 \le d \le n), \qquad 0 < t < 1, \tag{17}$$

so that every ratio $\varepsilon_d/\varepsilon_{d+1}$ equals t, and take $2\varepsilon_n < \delta$ (hence also $2\varepsilon_n < \delta_0$, as required in Lemma 7). By Lemma 9(1), every active set B for which (13) holds at some point has $|B| \le n$. For such B let $\lambda^B = (\lambda^B_a)_{a\in B}$, with values in $\bigoplus_{a\in B} L_a$, and let $P^\lambda_B = \lambda^B|_{T^{1,0}_I X}$, $Q^\lambda_B = \lambda^B|_{T^{0,1}_I X}$. By (16), P^λ_B differs from P_B in (11) by at most $2C_2\sqrt{n}\,t$, so for t smaller than a constant depending on C_0, C_2, n it is surjective with a right inverse of norm at most $2C_0$ (if $P' = P + \Delta$ with $\|\Delta\| \le 1/(2C_0)$, then $R(\mathrm{id} + \Delta R)^{-1}$ is a right inverse of P' of norm $\le 2C_0$). By (12) and (16),

$$\|Q^\lambda_B\| \le \sqrt{n}\,(2C_1\varepsilon_n + 2C_2 t) \le C_3(\varepsilon_n + t). \tag{18}$$

These bounds hold simultaneously for all such B at all points satisfying (13). Now first choose ε_n small and then t small so that $2C_0C_3(\varepsilon_n + t) < \mu_0$, where μ_0 is the constant of Lemma 6 for the fixed metrics. The thresholds are fixed from now on.

Proposition 10 (Local tame structures)**.** *Let B be an active set and x a point satisfying* (13)*. Then there are a neighbourhood $W \subset O_B$ of x and a smooth σ-tame almost complex structure on W for which all the functions f^B_a, $a \in B$, are holomorphic.*

Proof. Apply Lemma 6 with $V = \bigoplus_{a\in B} L_a$, $\lambda = \lambda^B$, $P = P^\lambda_B$, $Q = Q^\lambda_B$ and $R = P^*(PP^*)^{-1}$. At x we have $\|R\| \le 2C_0$ (some right inverse has norm $\le 2C_0$, and R has the least norm) and $\|Q\| \le C_3(\varepsilon_n + t)$, so $\|\mu\| < \mu_0$. Surjectivity of P and the bound on μ are open conditions, so they hold on a neighbourhood W of x, and the lemma provides J_μ on W; since $\lambda^B_a = F^B_a\, df^B_a$ with nonvanishing F^B_a, each f^B_a is J_μ-holomorphic. □

7 Simultaneous local choices and global patching

To patch the local structures of Proposition 10 we must make sure that near a point x *all* the functions f^A_a of *all* the sets U_A that come close to x are holomorphic for the local structure chosen at x. Lemma 8 reduces this to the functions of one maximal active set. For $x \in X$ let

$$B_x = \bigcup \{A :\ A \text{ active},\ x \in \overline{U_A}\}. \tag{19}$$

A union of ancestor-closed sets of non-leaf nodes is again one, so B_x is an active set ($B_x = \varnothing$ is allowed).

Lemma 11. *The point x satisfies the inequalities* (13) *for $B = B_x$. In particular $|B_x| \le n$ and the frames $F^{B_x}_c$ are defined near x.*

Proof. If $a \in B_x$, then $a \in A$ for some A with $x \in \overline{U_A}$, so $|s_a(x)| \le 2\varepsilon_{d(a)}$. For the boundary inequalities, let $b \in \partial B_x$. If b is a child of some $a \in B_x$ with $b \notin B_x$, choose A with $a \in A$ and

$x \in \overline{U_A}$; then $b \notin A$ because $A \subset B_x$, so $b \in \partial A$ and $|s_b(x)| \geq \varepsilon_{d(b)}$. If b is a root not in B_x, then b is a root not in A for every A occurring in (19), so $b \in \partial A$ and again $|s_b(x)| \geq \varepsilon_{d(b)}$. The remaining assertions follow from Lemma 9(1) and from $2\varepsilon_n < \delta$, as in Section 6. □

By Lemma 11 and Proposition 10 there are a neighbourhood $W_x \subset O_{B_x}$ of x and a σ-tame structure J_x on W_x for which every $f_a^{B_x}$, $a \in B_x$, is holomorphic (for $B_x = \varnothing$ take $J_x = I$). Shrink W_x so that

$$U_A \cap W_x \neq \varnothing \quad \Longrightarrow \quad A \subset B_x. \tag{20}$$

This is possible: there are finitely many active sets, and if $x \notin \overline{U_A}$ then some neighbourhood of x misses U_A, while $x \in \overline{U_A}$ gives $A \subset B_x$ by definition.

For every active A with $U_A \cap W_x \neq \varnothing$ we thus have $A \subset B_x$, and Lemma 8 expresses each f_a^A on $U_A \cap W_x \subset U_A \cap O_{B_x}$ as a Laurent monomial, with nonvanishing denominators, in the J_x-holomorphic functions $f_c^{B_x}$. Hence

$$f_a^A \text{ is } J_x\text{-holomorphic on } U_A \cap W_x \qquad (A \text{ active},\ a \in A). \tag{21}$$

Proposition 12 (The global structure)**.** *There is a σ-tame almost complex structure J on X such that*

$$df_a^A \circ J = i\, df_a^A \quad \text{on } U_A \qquad \text{for every active set } A \text{ and every } a \in A. \tag{22}$$

Proof. Choose finitely many points $x_1, \ldots, x_N$ with $X = \bigcup_k W_{x_k}$, a partition of unity (ρ_k) subordinate to this cover, and a σ-compatible reference structure K (for instance $K = I$, which is compatible). Let J be the tame structure of Lemma 5 with graph map $\Phi_J = \sum_k \rho_k \Phi_{J_{x_k}}$. Let $y \in U_A$ and $a \in A$. Every k with $\rho_k(y) \neq 0$ has $y \in U_A \cap W_{x_k}$, so f_a^A is J_{x_k}-holomorphic near y by (21). By Lemma 5, f_a^A is J-holomorphic at y. □

8 The coordinates and the deformation

Proof of Theorem 2. The coordinates. Let $p \in D$ and let $D_{i_1}, \ldots, D_{i_r}$ be the original branches through p; $1 \leq r \leq n$. By Lemma 7 choose an active set A with $p \in U_A$. Each root i_j belongs to A: a root not in A lies in ∂A and its section is nonzero on U_A, whereas $s_{i_j}(p) = 0$. Define $z_j = f_{i_j}^A$ for $1 \leq j \leq r$: smooth functions on U_A, J-holomorphic by (22), with $z_j^{-1}(0) = D_{i_j} \cap U_A$ because the frame $F_{i_j}^A$ does not vanish. At p, since $s_{i_j}(p) = 0$, $dz_j(p) = (F_{i_j}^A(p))^{-1} ds_{i_j}(p)$, and the real derivatives $ds_{i_1}(p), \ldots, ds_{i_r}(p)$ are jointly surjective onto $\bigoplus_j (L_{i_j})_p$ by the transversality of the branches through p (Definition 1). So $(z_1, \ldots, z_r)\colon U_A \to \mathbb{C}^r$ is a submersion at p, and the implicit function theorem completes it by smooth functions $z_{r+1}, \ldots, z_n$ to a smooth complex coordinate system on a neighbourhood of p. Shrink this neighbourhood so that it misses the branches not containing p. On it, $D = D_{i_1} \cup \cdots \cup D_{i_r} = \{z_1 \cdots z_r = 0\}$. For $p \notin D$ take a neighbourhood disjoint from D, arbitrary smooth complex coordinates and $r = 0$.

The deformation. The only changes of the symplectic form in the whole construction were the regularization deformations of Lemma 3, applied in Section 4: one inside each of the $m(2^n-1)$ applications of Lemma 4 and one at the end. Each is a path $\sigma + d\tau_t$ of forms cohomologous to its starting point along which every divisor constructed up to that stage, in particular every original branch D_i, remains part of a positive symplectic SNC collection. Concatenating these finitely many paths, reparametrized smoothly at the joins, gives the path ω_t of Theorem 2: $\omega_0 = \omega$, $[\omega_t] = [\omega]$, $\{D_i\}$ is positive SNC for every ω_t, and $\omega_1 = \sigma$. The structure J of Proposition 12 is σ-tame, that is, ω_1-tame. This completes the proof. □

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